

Math 465: Introduction to Combinatorics

Yelena Mandelshtam

Homework #3 is on Canvas

Early drop deadline on Monday! Homework 2 and quiz 2 will be graded before then.

These slides will be posted on Canvas.

Signless Stirling numbers of the first kind

Recall that $c(w)$ denotes the number of cycles in a permutation w .

Definition

The numbers

$$c(n, k) = \#\{w \in S_n \mid c(w) = k\} \quad (1 \leq k \leq n)$$

are called the signless Stirling numbers of the first kind.

There is no explicit closed formula for $c(n, k)$.

Corollary

$$\sum_{k=1}^n c(n, k)x^k = \sum_{w \in S_n} x^{c(w)} = x(x+1)(x+2) \cdots (x+n-1)$$

Recurrence for the signless Stirling numbers of the 1st kind

Proposition

$$c(n, k) = c(n-1, k-1) + (n-1)c(n-1, k)$$

Proof

The formula

$$\sum_{k=1}^n c(n, k)x^k = x(x+1)(x+2)\cdots(x+n-1)$$

implies the identity

$$\sum_{k=1}^n c(n, k)x^k = (x+n-1) \sum_{j=1}^{n-1} c(n-1, j)x^j.$$

Now compare the coefficients of x^k on both sides.

Exercise

Give a combinatorial proof of this proposition.

Triangle of the signless Stirling numbers of the first kind

The recurrence

$$c(n, k) = c(n-1, k-1) + (n-1)c(n-1, k)$$

can be used to recursively compute the Stirling numbers $c(n, k)$.

Triangle of the numbers $c(n, k)$ for $1 \leq k \leq n \leq 7$

						1							
					1		1						
			2			3		1					
		6		11		6		1					
	24		50		35		10		1				
120		274		225		85		15		1			
720		1764		1624		735		175		21		1	
...	...	...	...	...	...	...	...	...	...	...	...	...	...

The numbering of rows/columns starts with $n = 1$ (resp., $k = 1$).

Signed Stirling numbers of the first kind

Definition

The numbers

$$s(n, k) = (-1)^{n-k} c(n, k)$$

are called the **signed Stirling numbers of the first kind**.

				1						
			-1		1					
		2		-3		1				
	-6		11		-6		1			
24		-50		35		-10		1		
-120		274		-225		85		-15		1
...	...	...	...	...	...	...	...	...	...	...

Signed Stirling numbers of the first kind $s(n, k)$ for $n \leq 6$

Generating functions for the numbers $s(n, k)$

$$s(n, k) = (-1)^{n-k} c(n, k)$$

$$\sum_{k=1}^n c(n, k) x^k = x(x+1)(x+2) \cdots (x+n-1)$$

$$\begin{aligned} \sum_{k=1}^n s(n, k) x^k &= \sum_{k=1}^n (-1)^{n-k} c(n, k) x^k \\ &= (-1)^n \sum_{k=1}^n c(n, k) (-x)^k \\ &= (-1)^n (-x)(-x+1)(-x+2) \cdots (-x+n-1) \\ &= x(x-1)(x-2) \cdots (x-n+1) \end{aligned}$$

Expanding falling powers via ordinary powers

Corollary

$$\begin{aligned}\sum_{k=1}^n s(n, k)x^k &= x(x-1)(x-2)\cdots(x-n+1) \\ &= (x)_n \\ &= n! \binom{x}{n}\end{aligned}$$

Thus the Stirling numbers of the first kind arise when we express the falling powers in terms of ordinary powers.

Example: $n = 3$

$$2x - 3x^2 + x^3 = x(x-1)(x-2) = (x)_3.$$

Stirling numbers of the second kind

Definition

The **Stirling number of the second kind**, denoted $S(n, k)$, counts partitions of an n -element set into k nonempty subsets (**blocks**).

The blocks are not labeled. Alternative viewpoint: enumeration of equivalence relations on $\{1, \dots, n\}$ with k equivalence classes.

Example: $n = 4$

$S(4, 1) = 1$	$S(4, 2) = 7$		$S(4, 3) = 6$	$S(4, 4) = 1$																																					
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Recurrence for the Stirling numbers of the second kind

Theorem

$$S(n, k) = S(n - 1, k - 1) + k S(n - 1, k)$$

Proof

Each partition of $\{1, \dots, n\}$ into k blocks is of one of two kinds:

- either it has a block $\boxed{n}$ consisting solely of n , or
- it can be obtained by adding n to one of the blocks of a (unique) partition of $\{1, \dots, n-1\}$ into k blocks.

Table of the Stirling numbers of the second kind

The recurrence

$$S(n, k) = S(n-1, k-1) + k S(n-1, k)$$

can be used to compute the Stirling numbers of the second kind:

							1							
						1		1						
				1			3		1					
			1		7			6		1				
		1		15		25			10		1			
	1		31		90		65			15		1		
1		63		301		350		140			21		1	
..	..	..	..	..	..	..	..	..	..	..	..	..	..	..

Stirling numbers of the second kind $S(n, k)$ for $n \leq 7$

Expanding ordinary powers in terms of falling powers

Theorem

$$x^n = \sum_{k=1}^n S(n, k) (x)_k = \sum_{k=1}^n S(n, k) k! \binom{x}{k}$$

Thus, the Stirling numbers of the second kind arise when we expand a monomial x^n as a linear combination of the falling powers of x (or of binomial coefficients $\binom{x}{k}$).

Examples

$$x^2 = (x)_1 + (x)_2 = \binom{x}{1} + 2\binom{x}{2}$$

$$x^3 = (x)_1 + 3(x)_2 + (x)_3 = \binom{x}{1} + 6\binom{x}{2} + 6\binom{x}{3}.$$

Proof of the formula $x^n = \sum S(n, k) (x)_k$

Theorem

$$x^n = \sum_{k=1}^n S(n, k) (x)_k$$

Proof

It suffices to prove this identity when x is a positive integer [why?].
In that case,

x^n = number of ways to color $\{1, \dots, n\}$ in x colors;

$\sum_k S(n, k) (x)_k$ = number of ways to split $\{1, \dots, n\}$ into blocks,
then color these blocks in distinct colors.

Stirling numbers as entries of transition matrices

Consider the n -dimensional vector space

{polynomials in x of degree $\leq n$ with constant term 0}.

This vector space has two distinguished bases:

- the basis of monomials $(x, x^2, \dots, x^n)$;
- the basis of falling powers $((x)_1, (x)_2, \dots, (x)_n)$.

The Stirling numbers of the two kinds describe the transitions from one of these bases to another. Consequently, the (lower-triangular) matrices made of these numbers are inverse to each other.

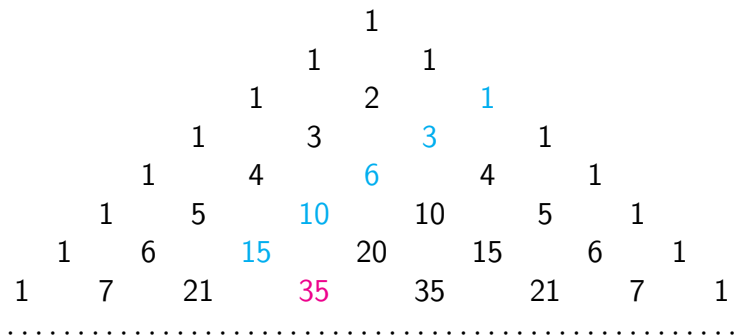
Example

$$\begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 2 & -3 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 3 & 1 \end{bmatrix}^{-1}.$$

Adding binomial coefficients: the hockey stick identity

Lemma

$$\sum_{j=0}^n \binom{j}{k} = \sum_{j=k}^n \binom{j}{k} = \binom{n+1}{k+1}$$



Adding the squares

Proposition

$$\sum_{j=1}^n j^2 = \frac{n(n+1)(2n+1)}{6}$$

Proof

$$\begin{aligned}\sum_{j=1}^n j^2 &= \sum_{j=1}^n \left(\binom{j}{1} + 2\binom{j}{2} \right) \\ &= \binom{n+1}{2} + 2\binom{n+1}{3} \\ &= \frac{(n+1)n}{2} + \frac{(n+1)n(n-1)}{3} \\ &= \frac{n(n+1)(2n+1)}{6}\end{aligned}$$

Summation formulas

More generally, expressing a polynomial $f(j)$ in terms of binomial coefficients $\binom{j}{k}$ leads to a summation formula for f :

$$\begin{aligned} f(j) &= \sum_k a_k \binom{j}{k} \\ \implies \sum_{j=0}^n f(j) &= \sum_{j=0}^n \sum_k a_k \binom{j}{k} \\ &= \sum_k a_k \binom{n+1}{k+1} \end{aligned}$$

Linear recurrences: motivating example

Problem

Solve the **linear recurrence**

$$h_n = 5h_{n-1} - 6h_{n-2} \quad (n \geq 2)$$

with the initial conditions $h_0 = 1$ and $h_1 = -2$.

Solution

$$\begin{aligned}\sum_{n \geq 2} h_n x^n &= 5 \sum_{n \geq 2} h_{n-1} x^n - 6 \sum_{n \geq 2} h_{n-2} x^n \\ &= 5x \sum_{n \geq 2} h_{n-1} x^{n-1} - 6x^2 \sum_{n \geq 2} h_{n-2} x^{n-2} \\ &= 5x \sum_{n \geq 1} h_n x^n - 6x^2 \sum_{n \geq 0} h_n x^n\end{aligned}$$

$$h(x) - h_1 x - h_0 = 5x(h(x) - h_0) - 6x^2 h(x)$$

Linear recurrences: the generating functions approach

Problem

Solve the recurrence relation

$$h_n = 5h_{n-1} - 6h_{n-2}$$

with the initial conditions $h_0 = 1$ and $h_1 = -2$.

Solution (continued)

$$h(x) - h_1x - h_0 = 5x(h(x) - h_0) - 6x^2h(x)$$

$$h(x) + 2x - 1 = 5x(h(x) - 1) - 6x^2h(x)$$

$$h(x) = \frac{1 - 7x}{1 - 5x + 6x^2} = \frac{1 - 7x}{(1 - 2x)(1 - 3x)} = \frac{5}{1 - 2x} - \frac{4}{1 - 3x}$$

$$h(x) = 5(1 + 2x + 4x^2 + 8x^3 + \dots) - 4(1 + 3x + 9x^2 + 27x^3 + \dots)$$

$$h_n = 5 \cdot 2^n - 4 \cdot 3^n \leftarrow \text{linear combination of geometric progressions}$$

While this approach can be developed systematically, we won't do it.

Linear recurrences: general set-up

We aim to solve **linear recurrences with constant coefficients**, i.e., recurrences of the form

$$(*) \quad h_n + a_1 h_{n-1} + a_2 h_{n-2} + \cdots + a_k h_{n-k} = 0.$$

Here

- k is the **order** of the recurrence;
- $a_1, \dots, a_k$ are its **coefficients**;
- $a_k \neq 0$ without loss of generality;
- the k **initial values** $h_0, h_1, \dots, h_{k-1}$ are given;
- n takes values $k, k+1, k+2, \dots$.

Example

$$h_n - 5h_{n-1} + 6h_{n-2} = 0$$

The theory of linear recurrences is very similar to the theory of linear differential equations.

The characteristic equation

$$(*) \quad h_n + a_1 h_{n-1} + a_2 h_{n-2} + \cdots + a_k h_{n-k} = 0$$

Lemma

For $q \neq 0$, the geometric progression $h_n = q^n$ satisfies $(*)$ if and only if

$$(**) \quad q^k + a_1 q^{k-1} + a_2 q^{k-2} + \cdots + a_k = 0.$$

Proof

$$\begin{aligned} q^n + a_1 q^{n-1} + a_2 q^{n-2} + \cdots + a_k q^{n-k} &= 0 \\ \iff q^k + a_1 q^{k-1} + a_2 q^{k-2} + \cdots + a_k &= 0 \end{aligned}$$

Equation $(**)$ is the **characteristic equation** for the recurrence $(*)$.

Example

$$(*) \quad h_n - 5h_{n-1} + 6h_{n-2} = 0$$

$$(**) \quad q^2 - 5q + 6 = 0$$

Roots: $q = 2$ and $q = 3$. Solutions $h_n = 2^n$ and $h_n = 3^n$.

The vector space of solutions of a linear recurrence

Lemma

The solutions of a linear recurrence

$$(*) \quad h_n + a_1 h_{n-1} + a_2 h_{n-2} + \cdots + a_k h_{n-k} = 0$$

form a vector space. The dimension of this vector space is k .

Example

$$(*) \quad h_n - 5h_{n-1} + 6h_{n-2} = 0$$

1	0	-6	-30	-114	...
0	1	5	19	65	...
a	b	$-6a+5b$	$-30a+19b$	$-114a+65b$	...

Geometric progressions

Lemma

Geometric progressions with distinct ratios are linearly independent.

Proof [sketch]

Suppose that k geometric progressions (in the example below, $k = 3$)

$$\begin{array}{cccc} 1 & q & q^2 & \cdots \\ 1 & r & r^2 & \cdots \\ 1 & s & s^2 & \cdots \end{array}$$

(with q, r, s distinct and nonzero) are linearly dependent. Then

$$\det \begin{pmatrix} 1 & q & q^2 \\ 1 & r & r^2 \\ 1 & s & s^2 \end{pmatrix} = 0 \Rightarrow \exists (a, b, c) \neq (0, 0, 0) \begin{cases} a + bq + cq^2 = 0, \\ a + br + cr^2 = 0, \\ a + bs + cs^2 = 0. \end{cases}$$

Thus $a + bt + ct^2$, a nonzero polynomial of degree $\leq k - 1$, has k distinct roots, a contradiction.

General solution in the case of distinct roots

$$(*) \quad h_n + a_1 h_{n-1} + a_2 h_{n-2} + \cdots + a_k h_{n-k} = 0$$

Three Lemmas

- The solutions of $(*)$ form a vector space of dimension k .
- A geometric progression $h_n = q^n$ satisfies $(*)$ if and only if

$$(**) \quad q^k + a_1 q^{k-1} + a_2 q^{k-2} + \cdots + a_k = 0.$$

- Geometric progressions with distinct ratios are linearly independent.

These lemmas imply the following theorem.

Theorem

If the characteristic equation $(**)$ has distinct roots $q_1, \dots, q_k$, then

$$h_n = \sum_{i=1}^k c_i q_i^n$$

is the general solution of $(*)$.

The case of distinct roots: example

Theorem

Given a linear recurrence

$$(*) \quad h_n + a_1 h_{n-1} + a_2 h_{n-2} + \cdots + a_k h_{n-k} = 0,$$

if the characteristic equation

$$q^k + a_1 q^{k-1} + a_2 q^{k-2} + \cdots + a_k = 0$$

has distinct roots $q_1, \dots, q_k$, then the general solution of $(*)$ is given by

$$h_n = \sum_{i=1}^k c_i q_i^n.$$

Example

The general solution of the recurrence $h_n - 5h_{n-1} + 6h_{n-2} = 0$ is given by $h_n = c_1 \cdot 2^n + c_2 \cdot 3^n$. For the initial conditions $h_0 = 1$ and $h_1 = -2$, we get $c_1 + c_2 = 1$ and $2c_1 + 3c_2 = -2$, implying $c_1 = 5$ and $c_2 = -4$.